

Hate Speech Detection

A mini NLP project using ETHOS, TF-IDF and Logistic Regression

BUILD • AUDIT • EXPLAIN

PROJECT TEAM

Team members

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Dataset: ETHOS Hate Speech Dataset

Task: binary classification of English online comments

The problem we are solving

Automated hate-speech detection can help prioritize comments for human review.

PROBLEM STATEMENT

Large volumes of social-media text make manual review slow and difficult.

The challenge is not only recognizing offensive words. Hate speech may depend on the target, wording and context, while profanity can also appear in non-hateful comments.

PROJECT OBJECTIVES

- Build an end-to-end NLP classifier using a real dataset.
- Measure performance using multiple evaluation metrics.
- Audit two text styles and explain the features behind predictions.

OUR APPROACH

01

Load ETHOS

Real online comments

02

Train baseline

TF-IDF + Logistic Regression

03

Audit results

Style groups + feature weights

Human review remains essential.

Dataset: ETHOS Hate Speech

The binary dataset contains 998 English-language online comments.

998**Total comments**

Real online text

433**Hate**

43.4% of data

565**Non-Hate**

56.6% of data

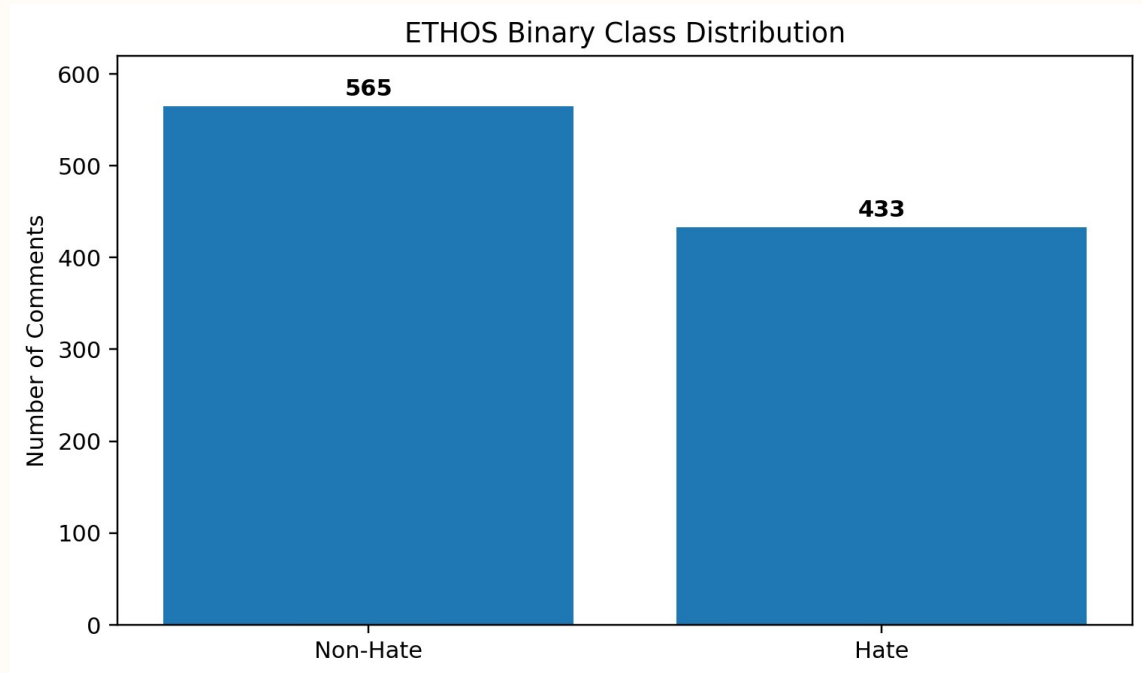
LABEL RULE

isHate ≥ 0.5 → Hate (1)

isHate < 0.5 → Non-Hate (0)

- 80/20 stratified train/test split
- 798 training comments • 200 test comments

OUTPUT · CLASS DISTRIBUTION



Notebook output: comments per class

Source: ETHOS dataset (Kaggle). Label conversion uses the 0.5 threshold described in the project.

From raw text to prediction

A simple, interpretable supervised-learning pipeline.



FEATURE REPRESENTATION · TF-IDF

- Turns text into a sparse numerical vector.
- Unigrams capture words; bigrams capture two-word phrases.

CLASSIFIER · LOGISTIC REGRESSION

- Learns coefficients for the TF-IDF features.
- Balanced class weights; max_iter=2000; random_state=42.

Tools: Python · Pandas · NumPy · scikit-learn · Matplotlib · Seaborn · KaggleHub

Overall test-set performance

Metrics are measured on 200 held-out comments.



CLASS-SPECIFIC SCORES

Metric	Hate	Non-Hate
Precision	0.676	0.706
Recall	0.575	0.788
F1-score	0.621	0.745

INTERPRETATION

69.5% vs. 56.5% majority-class baseline

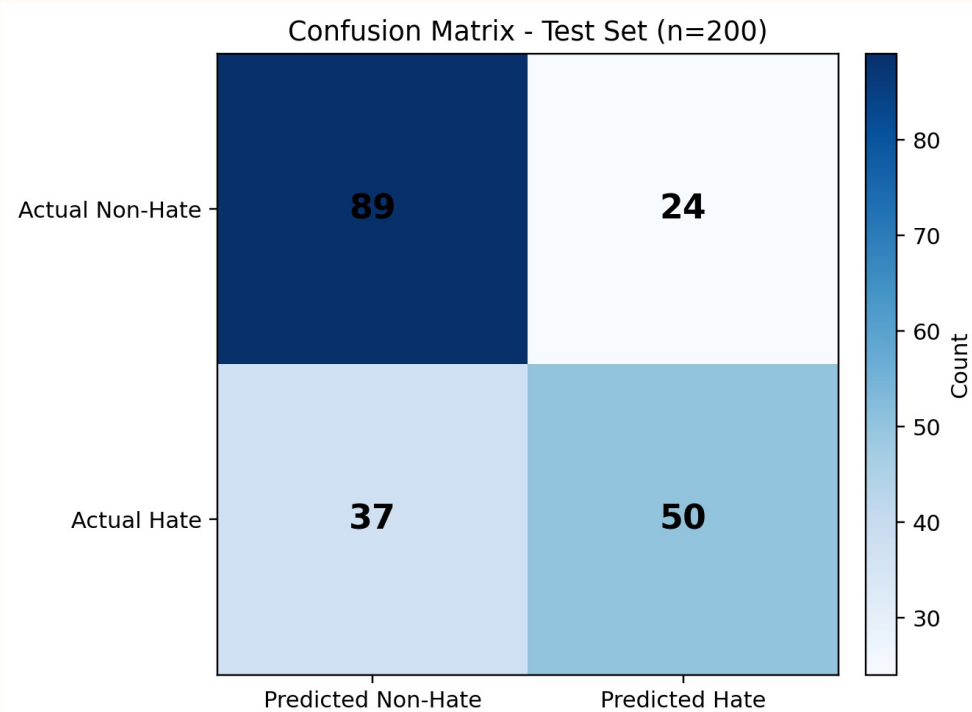
The model improves on always predicting Non-Hate, but hate recall remains limited: it misses 37 of 87 hate comments.

Metrics use the same fixed stratified split (random_state=42); no cross-validation or hyperparameter tuning was performed.

What the predictions got right — and wrong

The confusion matrix separates correct predictions from two kinds of error.

OUTPUT · CONFUSION MATRIX



Notebook output: confusion matrix on the 200-comment test set

READING THE MATRIX

89

True non-hate

Correctly left unflagged

50

True hate

Correctly detected

24

False positives

Non-hate flagged as hate

37

False negatives

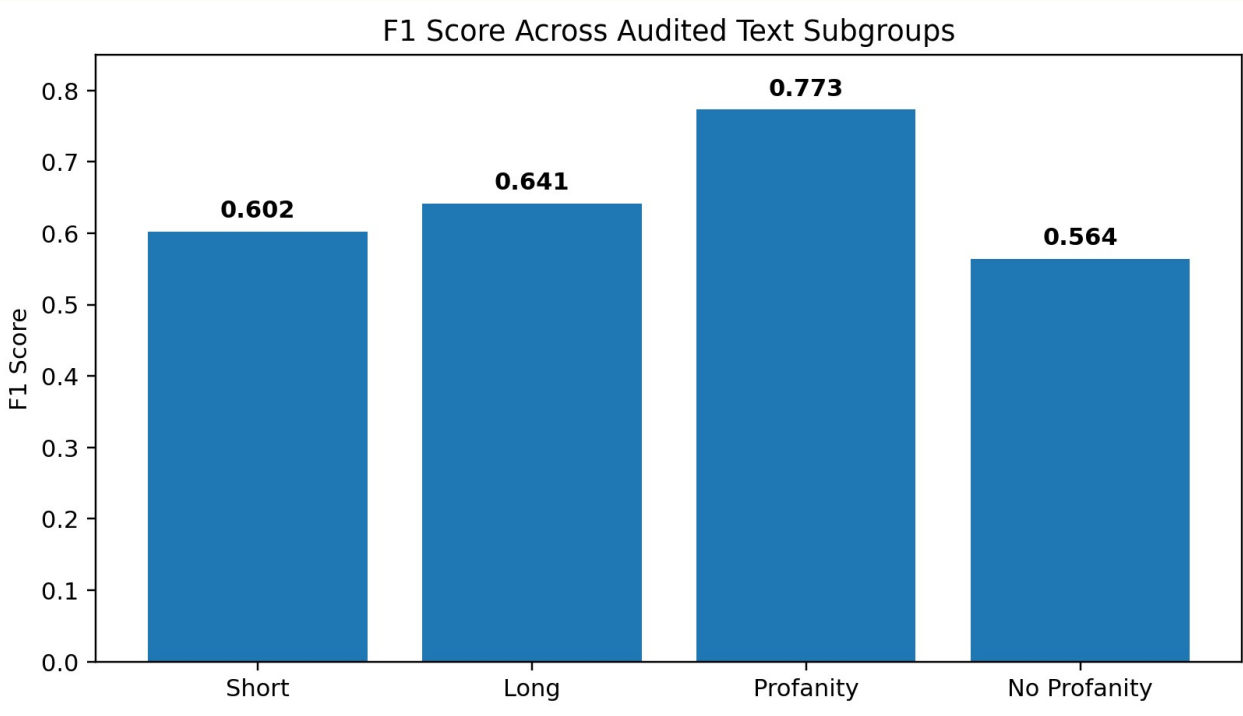
Hate missed by the model

The model misses 42.5% of hate comments.

Performance changes by text style

We compared short vs. long comments and profanity vs. no profanity.

● ● ● OUTPUT · SUBGROUP AUDIT



Notebook output: F1-score for four test-set subgroups

The profanity subgroup contains only 31 comments, so the result is a warning signal—not a definitive fairness estimate.

AUDIT SNAPSHOT

Group	Hate recall	F1
Short	59.5%	60.2%
Long	55.6%	64.1%
Profanity	89.5%	77.3%
No profanity	48.5%	56.4%

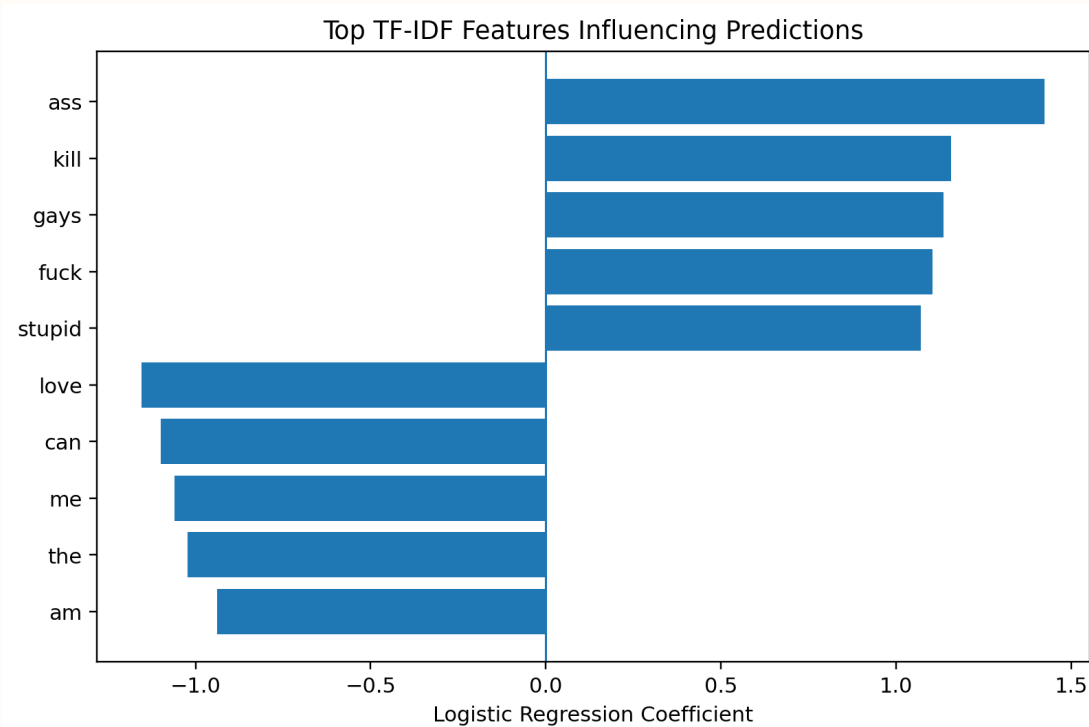
Key finding

Hate recall drops from 89.5% with profanity to 48.5% without it.

Which words influence the model?

A simple coefficient inspection explains the direction of learned feature influence.

OUTPUT · TOP FEATURES



Notebook output: top positive and negative TF-IDF coefficients

HOW TO READ THIS

Positive coefficient

Pushes prediction toward Hate

Negative coefficient

Pushes prediction toward Non-Hate

EXAMPLES FROM THE FITTED MODEL

Hate-leaning ass · kill · gays · fuck · stupid

Non-hate-leaning love · can · me · the · am

These are learned associations, not causal explanations.
Identity terms may be flagged in harmless contexts.

Failure cases and limitations

The model detects lexical patterns, but not full meaning or intent.

ILLUSTRATIVE MISS

“People from that group should leave”

Predicted: Non-Hate · hate probability 0.36

Exclusionary meaning may be missed when lexical signals are weak.

ILLUSTRATIVE FALSE FLAG

“I disagree with your opinion”

Predicted: Hate · hate probability 0.57

A neutral statement may be flagged because of learned word associations.

MAIN LIMITATIONS

- | | | | |
|--|---|--|--|
|  Small dataset | 998 comments; only 200 held out for testing. |  Limited context | Bag-of-words features do not understand sarcasm, intent or conversation. |
|  Shortcut risk | Profanity and identity-related words may trigger incorrect predictions. |  Narrow scope | English-only data; no cross-validation or model tuning. |

The two example predictions are hand-written demonstrations, not examples from the held-out test set.

Conclusion and next steps

A complete baseline pipeline—plus an audit that reveals where it breaks.

WHAT WE LEARNED

69.5%

test accuracy

- Built a reproducible TF-IDF + Logistic Regression pipeline.
- Measured performance by subgroup, not just overall accuracy.
- Found a large recall gap related to profanity usage.

Use for first-pass support with human review—not automatic moderation.

FUTURE IMPROVEMENTS

- Use cross-validation and tune the decision threshold on validation data.
- Compare Linear SVM or contextual models such as BERT.
- Expand the audit and test neutral identity-term mentions.

SOURCES

ETHOS Hate Speech Dataset (Kaggle)
Mollas et al. (2020), “ETHOS: an Online Hate Speech Detection Dataset”
scikit-learn documentation: TfidfVectorizer and LogisticRegression